Skills Summary

Three Errors in Parametric, Polar, and Vector Calculus

Use this reference while completing your worksheet or when studying.

Skill 1 — Derivatives of a parametrised curve

Slope: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$ — never dy/dt on its own. **Concavity:** differentiate that slope with respect to t and divide by dx/dt *again*:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}.$$

The second division is the step almost everyone drops.

Example: $x = t^3$, $y = t^2 - t$. Find $\dfrac{dy}{dx}$ at $t = 2$.

Step 1. The curve is given through a parameter, so the slope is a ratio of the two t-derivatives rather than either one alone.

Step 2. $\dfrac{dy}{dt} = 2t - 1$ and $\dfrac{dx}{dt} = 3t^2$, so at $t = 2$ the ratio is $\dfrac{3}{12}$.

$\Rightarrow$ **0.25**

Try it: $x = t^3$, $y = t^2 + t$. Find $\dfrac{dy}{dx}$ at $t = 2$.

check: **0.417**

Skill 2 — Polar bounds that trace the region once

Area: $A = \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta} r^2\, d\theta$, where $[\alpha, \beta]$ traces the boundary *exactly once*. Find the bounds from the equation, by solving $r = 0$ and taking two consecutive solutions. Circles $r = a\sin\theta$ and $r = a\cos\theta$ close after an interval of length π, not 2π; a petal of $r = a\cos(n\theta)$ runs from $-\frac{\pi}{2n}$ to $\frac{\pi}{2n}$.

Example: Find the area enclosed by $r = 6\cos\theta$.

Step 1. This is a circle, so the first job is the interval that draws it once: solving $6\cos\theta = 0$ gives the endpoints.

Step 2. The zeros are $\theta = \pm\frac{\pi}{2}$, so $A = \frac{1}{2}\int_{-\pi/2}^{\pi/2} 36\cos^2\theta\, d\theta = 18 \cdot \frac{\pi}{2}$.

$\Rightarrow$ $\boldsymbol{A = 9\pi \approx 28.274}$

Try it: Find the area enclosed by $r = 4\cos\theta$.

check: $\boldsymbol{A = 4\pi \approx 12.566}$

Skill 3 — Speed is a length, not a sum

Speed at time t is $\sqrt{\big(x'(t)\big)^2 + \big(y'(t)\big)^2}$ — the length of the velocity vector, computed from the *derivatives*, never from the position. **Distance travelled** is $\int_a^b \text{speed}\, dt$; a square root cannot be split across the sum inside it. To decide speeding up or slowing down, take the sign of $v \cdot a$.

Example: A particle has $x'(t) = 3$ and $y'(t) = 4t$. Find its speed at $t = 2$.
Step 1. Speed asks for the length of the velocity vector, so evaluate both derivatives at the instant and then take the magnitude.
Step 2. At $t = 2$ the velocity is $\langle 3, 8 \rangle$, so the speed is $\sqrt{9 + 64} = \sqrt{73}$.
$\Rightarrow$ **8.544**

Try it: A particle has $x'(t) = 5$ and $y'(t) = 3t$. Find its speed at $t = 4$.

check: **13**